JUAN MAC DONAGH

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17/09/1996 Address: 63 n°81, CP 1900, La Plata, Buenos Aires
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Profile

PhD candidate in structural bioinformatics (CONICET – National University of Quilmes). My thesis characterizes the structure-function diversity of the human proteome, studying how different types of proteins (ordered, disordered, repetitive, phase separating) differ in their evolutionary rates and conformational diversity. As part of this work, I study how transposable elements generate protein disorder in proteins specific to primates. I combine evolutionary and structural analysis with programming and ML tools, and I am actively involved in teaching, open science and the scientific community.

Graduate and Postgraduate Education

National Scientific and Technical Research Council (CONICET) - National University of Quilmes, Argentina:
PhD fellow at the Structural Bioinformatics Group (SBG). Advisor: Prof. Dr. Gustavo Parisi **2022 – 2027**
Thesis: *Characterization of structure-function diversity in the human structurome.*
Expected thesis defense: late February / early March 2027.

Faculty of Exact Sciences - National University of La Plata:
Bachelor's Degree in Biotechnology and Molecular Biology. GPA: 7.68/10, final thesis grade: 10/10 (Argentine 1–10 scale). **2016 – 2021**

Research stays and international experience

- **Johannes Gutenberg-Universität Mainz, Computational Biology and Data Mining Group (Prof. Miguel A. Andrade-Navarro)**
  Marie Curie Fellow – Horizon 2020 MSCA-RISE "REFRACT": 11/2022 – 01/2023
  Output: first author paper on structured tandem repeats in protein interactions (*IJMS*, 2024).
  Marie Curie Fellow – Horizon 2020 MSCA-RISE "REFRACT": 03/2024 – 05/2024
  Marie Curie Fellow – Horizon Europe MSCA Staff Exchanges "IDPfun2": 03/2025 – 08/2025
  Output: Coconut, a C suite for transcript, codon usage and protein structure analysis (in development).
- **European Molecular Biology Laboratory (EMBL), Heidelberg – Gibson Lab**
  Marie Curie Fellow – Horizon 2020 MSCA-RISE "IDPfun": 02/2023 – 05/2023
  Output: contribution to the ELM 2024 update (*Nucleic Acids Research*, 2024).

Publications

[*] **Juan Mac Donagh**, Nicolas Vergesio, Andrea Aguilar, Rodrigo Nores, Antonio Lagares, Maria Silvina Fornasari, and Gustavo Parisi. *Transposable elements as evolutionary substrates of protein disorder in the human proteome.* Preprint. 2026. DOI: 10.64898/2026.06.12.731867.

[*] **Juan Mac Donagh**, Nahuel Escobedo, Tadeo Saldaño, Luciana Rodriguez Sawicki, Nicolas Palopoli, Sebastian Fernandez Alberti, Maria Silvina Fornasari, and Gustavo Parisi. "Revealing Missing Protein–Ligand Interactions Using AlphaFold Predictions". In: *Journal of Molecular Biology* (2024). DOI: 10.1016/j.jmb.2024.168852.

[*] **Juan Mac Donagh**, Abril Marchesini, Agostina Spiga, Maximiliano José Fallico, Paula Nazarena Arrías, Alexander Miguel Monzon, Aimilia-Christina Vagiona, Mariane Gonçalves-Kulik, Pablo Mier, and Miguel A. Andrade-Navarro. "Structured Tandem Repeats in Protein Interactions". In: *International Journal of Molecular Sciences* (2024). DOI: 10.3390/ijms25052994.

[*] Manjeet Kumar, Sushama Michael, Jesús Alvarado-Valverde, András Zeke, Tamas Lazar, Juliana Glavina, Eszter Nagy-Kanta, **Juan Mac Donagh**, Zsofia E Kalman, Stefano Pascarelli, Nicolas Palopoli, László Dobson, Carmen Florencia Suarez, Kim Van Roey, Izabella Krystkowiak, Juan Esteban Griffin, Anurag Nagpal, Rajesh Bhardwaj, Francesca Diella, Bálint Mészáros, Kellie Dean, Norman E Davey, Rita Pancsa, Lucía B Chemes, and Toby J Gibson. "ELM—the Eukaryotic Linear Motif resource—2024 update". In: *Nucleic Acids Research* (2024). DOI: 10.1093/nar/gkad1058.

[*] **Juan Mac Donagh**, Diego Javier Zea, Guillermo Benitez, Cristian Guisande Donadio, Julia Marchetti, Nicolas Palopoli, María Silvina Fornasari, and Gustavo Parisi. *Evolutionary rates in human amyloid proteins reveal their intrinsic metastability.* Preprint. 2022. DOI: 10.1101/2022.09.07.506994.

[*] Tadeo Saldaño, Nahuel Escobedo, Julia Marchetti, Diego Javier Zea, **Juan Mac Donagh**, Ana Julia Velez Rueda, Eduardo Gonik, Agustina García Melani, Julieta Novomisky Nechcoff, Martín N. Salas, Tomás Peters, Nicolás Demitroff, Sebastian Fernandez Alberti, Nicolas Palopoli, Maria Silvina Fornasari, and Gustavo Parisi. "Impact of protein conformational diversity on AlphaFold predictions". In: *Bioinformatics* (2022). DOI: 10.1093/bioinformatics/btac202.

Conference Presentations

[*] **Juan Mac Donagh** and Gustavo Parisi. "Transposable Element Insertions Contribute to the Emergence of Protein Disorder during Primate Evolution". In: *IDPfun2 Midterm Meeting / JIF UNQ*. Main talk. 2025.

[*] **Juan Mac Donagh**, Miguel A. Andrade-Navarro, and Gustavo Parisi. "Identifying differential codon usage bias in protein domains with an automated pipeline." In: *XIV A2B2C 2024*. Poster. 2024.

[*] **Juan Mac Donagh**, Diego Zea, and Gustavo Parisi. "Proteins that adopt condensate states occupy extreme and opposing positions within the spectrum of evolutionary rates in the human proteome". In: *XIV A2B2C 2024*. Poster. 2024.

[*] **Juan Mac Donagh**, Diego Zea, Nicolas Palopoli, María Silvina Fornasari, and Gustavo Parisi. "Evolutionary analysis reveals emerging properties of the different forms that proteins can adopt". In: *RABE V*. Poster. 2023.

[*] **Juan Mac Donagh**, Diego Zea, Gustavo Benítez, C.E. Guisande Donadio, J. Marchetti, Nicolas Palopoli, María Silvina Fornasari, and Gustavo Parisi. "Evolutionary rates in human amyloid proteins reveal their intrinsic metastability". In: *XII A2B2C 2022*. Poster. 2022.

Fellowships and Funding

CONICET Doctoral Fellowship, National Scientific and Technical Research Council, Argentina. **2022 – 2027**

Participation as team member in funded projects:

- Horizon Europe MSCA Staff Exchanges "IDPfun2" (GA 101182949). 2025 – 2029
- PICT-2022: "Amyloidogenic proteins: an evolutionary and population-based approach to elucidate the sequence bases of their metastability". 2024 – present
- ORFG Open Scholarship Seed Award: "Open training material in Spanish for database curation". 02/2024
- PICT-2018: "Evolutionary trajectories, conformational diversity and structural divergence as foundations for the development of bioinformatics tools for protein study". 2018 – 2024
- Horizon 2020 MSCA-RISE "IDPfun". 2017 – 2022
- Priority Research Program of the National University of Quilmes: "Simulation of molecular processes of physicochemical and biological relevance".

Teaching and Supervision

Teaching Assistant in Introduction to Chemistry and General Chemistry
National University of La Plata, Faculty of Exact Sciences. **03/2021 – present**

Trainer, 1st IDPfun2 Training School – *Insights into the molecular mechanisms of Intrinsically Disordered Proteins and their role in diseases: concepts, tools, and applications*
Horizon Europe MSCA Staff Exchanges, organized by the University of Padova (online). Co-led the practical session "Molecular specificity in IDR-mediated interactions (SLiMs context)" with Dr. Lucía Chemes. **04/2026**

Co-advisor and tutor, undergraduate final thesis of Facundo Sanchez Trapes
National University of La Plata, Faculty of Exact Sciences. **2023**

Leadership experience

Co-founder and vice president of the Young Researchers Association (AJI). **2024 – present**
Organizer, 2nd AJI Conference: *More! Federal science at the end of the world* (National University of San Martín, UNSAM, Buenos Aires, Argentina). **2026**
Organizer, 1st AJI Conference: *Federal science at the end of the world* (Fundación Instituto Leloir, Buenos Aires, Argentina). **10/2025**

Scientific Community Involvement

Volunteer, organizing team, XV Congress of the Argentine Association for Bioinformatics and Computational Biology (A2B2C, Buenos Aires, Argentina). **11/2026**
Volunteer, XIV Congress of the Argentine Association for Bioinformatics and Computational Biology (A2B2C, La Plata, Argentina). **11/2024**
Volunteer, Argentine Regional Student Group in Bioinformatics (RSG). **2022 – 2023**

Conference attendance

1st ML4NGP Training School: *Protein aggregation, intrinsic disorder and phase separation in the era of ML* (Porto, Portugal). **04/2023**
EMBO Workshop VIZBI: *Visualizing biological data* (EMBL, Heidelberg, Germany). **03/2023**
SHE2Bioinfo: Symposium of Spanish-speaking Bioinformatics and Computational Biology Students (virtual). **06/2022**

Skills and Competencies

Programming languages: Extensive experience: R, Python, Julia, Bash. Intermediate experience: C, Unity.
Structural bioinformatics: AlphaFold2/3, ChimeraX, PyMOL, TM-score (structure prediction, comparison, analysis and visualization).
Sequence & evolution: multiple sequence alignment (MAFFT, MUSCLE); homology and profile search (BLAST suite, HMMER, HH-suite); evolutionary rates (Rate4Site); UCSC Genome Browser.
Intrinsic disorder: IUPred, MobiDB, PUNCH2-light.
Machine / deep learning: PyTorch, TensorFlow/Keras.
R ecosystem: tidyverse, Bioconductor (Biostrings, GenomicRanges), bio3d, ape, phangorn, RMarkdown/Quarto, Shiny.
Development & infrastructure: Git, Docker, LaTeX, HTML/CSS/JS, computing clusters and HPC, reproducible workflows (Snakemake).
Data curation & open science: ELM (Eukaryotic Linear Motif resource); DisProt.
Languages: English (near-native; TOEFL, 2022); Spanish (native); Portuguese (intermediate); German (basic).

References

Prof. Dr. Gustavo Parisi – National University of Quilmes, Argentina (PhD advisor). gusparisi@gmail.com
Prof. Dr. Miguel A. Andrade-Navarro – Johannes Gutenberg-Universität Mainz, Germany. andrade@uni-mainz.de
Dr. Toby J. Gibson – EMBL Heidelberg, Germany. toby.gibson@embl.de